Testing the impact of cash transfers on youth using ABCD

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Cash transfers could improve mental health

- Socioeconomic disadvantage compounds risk
  - Adolescents from the lowest incomes are 3 times as likely to have mental health problems[1]
- Cash transfers show substantial, positive impacts on mental health, mostly in adults[2]
- Small, significant effect on adolescent mental health[3]:
  - small set of studies
  - costly RCTs
  - very few studies test impact of economic policy
- **ABCD provides an opportunity to capture development in the context of real-world economic policies**

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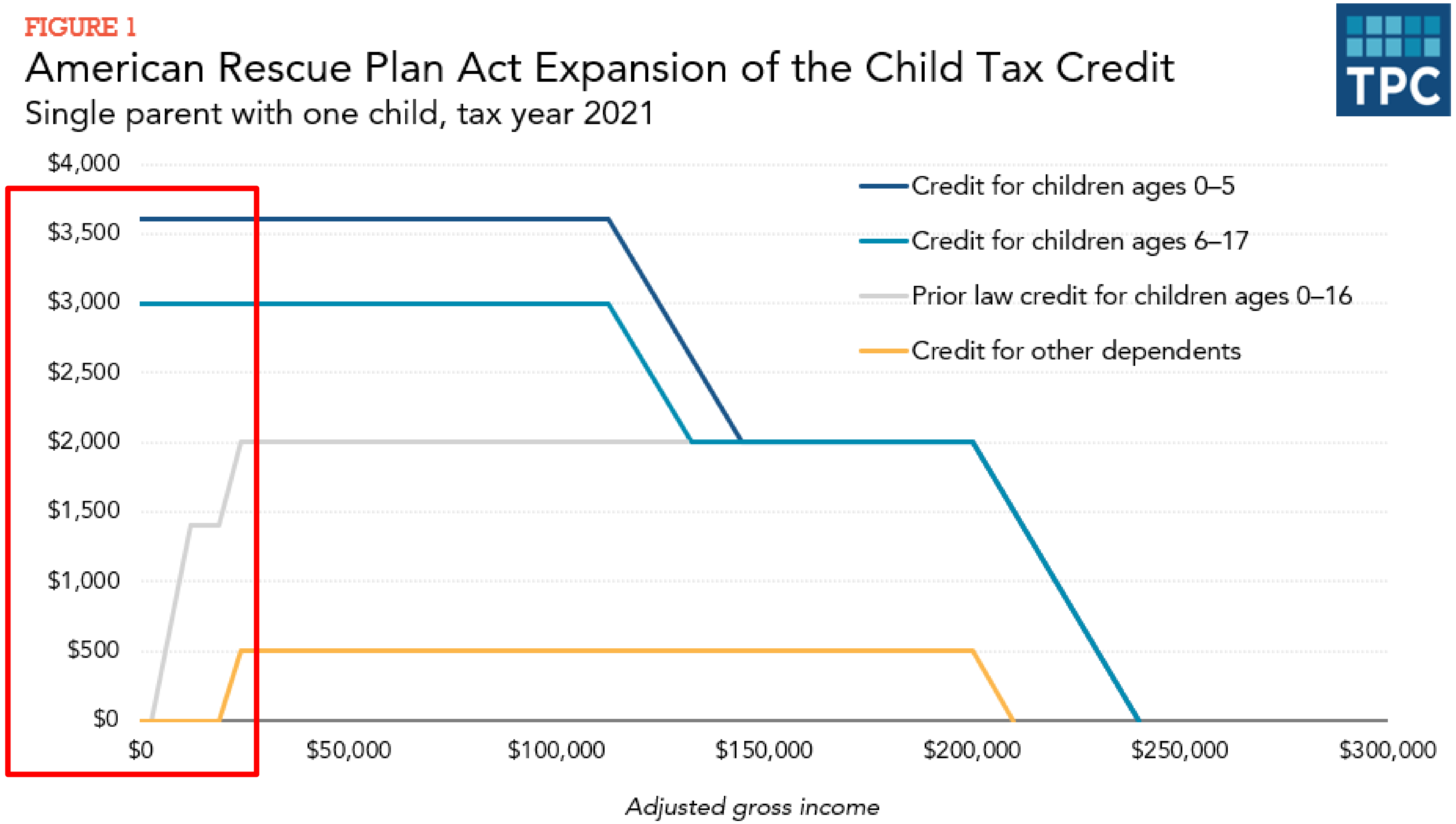
1. Reiss (2013), *Soc. Sci. Med.* 2. Ridley (2024) *Economics.* 3. Jaffee (2025) *Child Psych. Psy.*

Do cash transfers reduce mental health issues for adolescents in lowest income groups?

Build a natural experiment using observational data

Do cash transfers improve affordability for adolescents in lowest income groups?

- Expansion reached 61 million children
- First payment alone reduced child poverty rate by 24%
- 19 million under 17s in the lowest income bracket were now eligible for CTC
- This reverted in 2022



Source: Urban-Brookings Tax Policy Center calculations.
Notes: ARP = American Rescue Plan Act. Assumes all income comes from earnings, and child meets all tests to be a CTC-qualifying dependent. $3,000 and $3,600 credits are fully refundable; prior law limited refunds to $1,400 out of the maximum $2,000 credit. Credit for married parents first phases out at $150,000 of income until credit reaches pre-2021 level; begins second phase out at $400,000 of income. Only citizen children qualify for the $3,000 and $3,600 credits for children under 18. Noncitizens under age 18 who meet the dependency tests of eligibility can qualify other dependent credit.

figure source: taxpolicycenter.org
ref: Curran (2024) *AAPSS*

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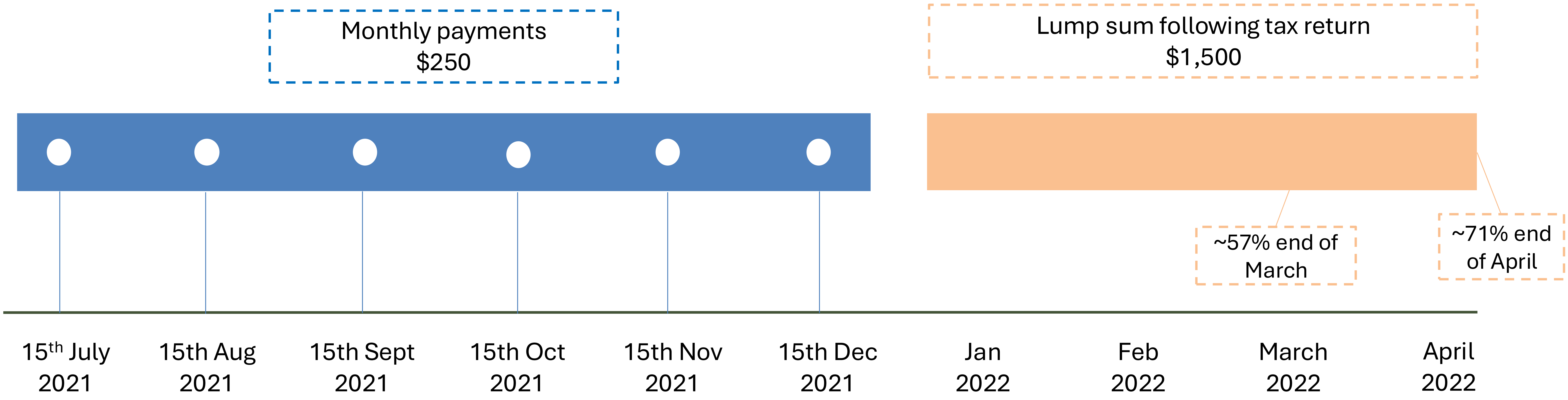
American Rescue Plan Act 2021: Expanded Child Tax Credit



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Affordability index in ABCD

Demographics questionnaire, 7 items, summed: higher score = greater affordability

In the past 12 months, has there been a time when you and your immediate family experienced any of the following:

1. Needed food but couldn't afford to buy it or couldn't afford to go out to get it?
2. Were without telephone service because you could not afford it?
3. Didn't pay the full amount of the rent or mortgage because you could not afford it?
4. Were evicted from your home for not paying the rent or mortgage?
5. Had services turned off by the gas or electric company, or the oil company wouldn't deliver oil because payments were not made?
6. Had someone who needed to see a doctor or go to the hospital but didn't go because you could not afford it?
7. Had someone who needed a dentist but couldn't go because you could not afford it?

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Methods & RQs

Difference-in-differences design

- Compares the *change* in affordability before vs. after expanded CTC between eligible and comparison groups allowing us to isolate the policy effect

$$\mathrm{DiD} = (\bar{Y}_{\text{Treatment, Post}} - \bar{Y}_{\text{Treatment, Pre}}) - (\bar{Y}_{\text{Control, Post}} - \bar{Y}_{\text{Control, Pre}})$$

Research questions:

- Did the monthly expanded CTC payments (July-December 2021) improve reported affordability for the lowest income families, relative to the comparison groups?
- *Income <$35k: N = 1103, Income $35k-$75k: N = 1239, Age = 13.5*

- Did any improvement in affordability last beyond the period of active monthly payments (July to December 2021)?
- *Income <$35k: N = 699, Income $35k-$75k: N = 509, Age = 13.5*

Group average treatment effect for multiple time periods: $ATT(g,t) = E[Y_t(g) - Y_t(0) | G = g]$

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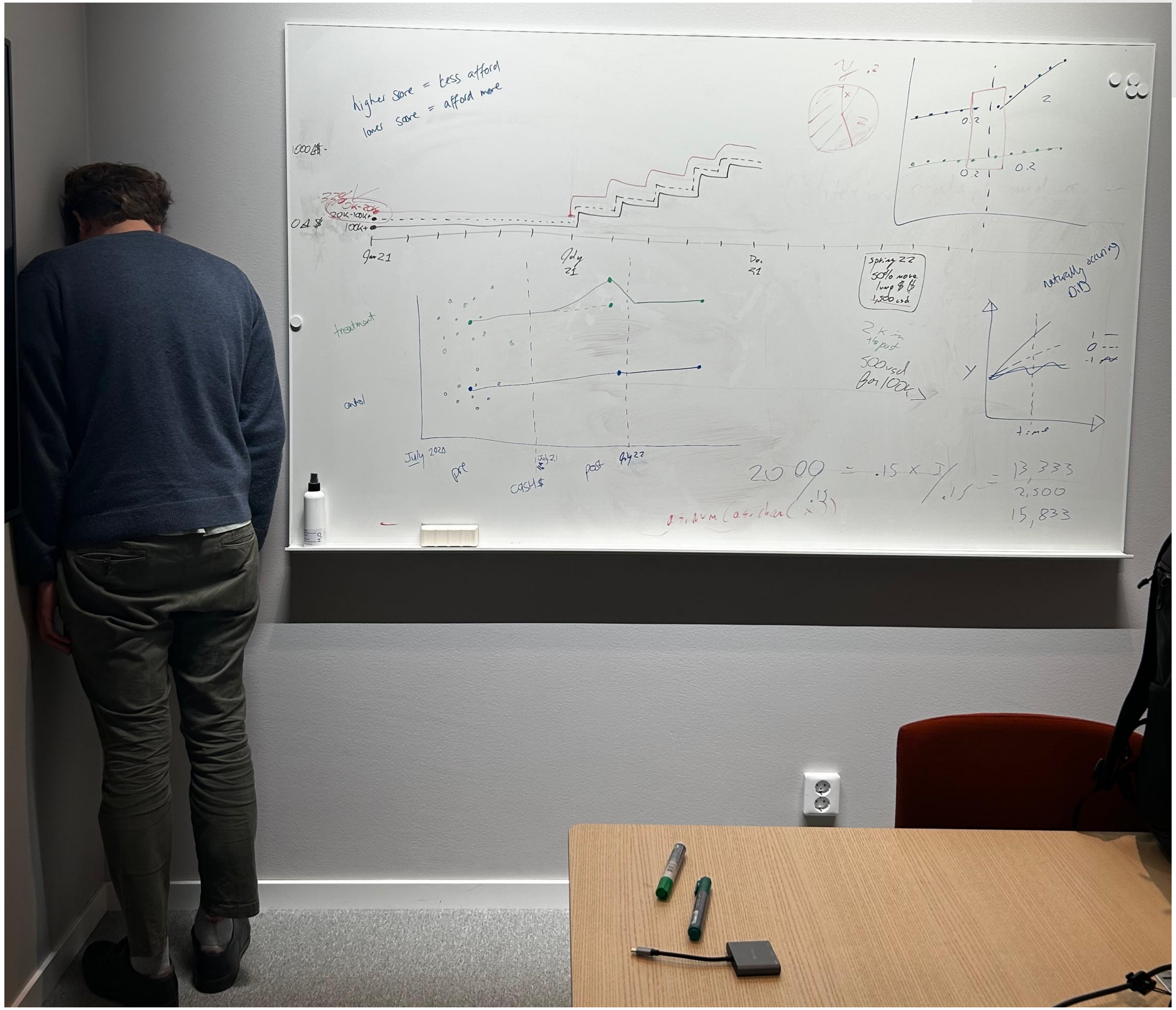
DinD R package, Callaway & Sant'Anna: https://bcallaway11.github.io/did/index.html

Addressing assumptions in DinD



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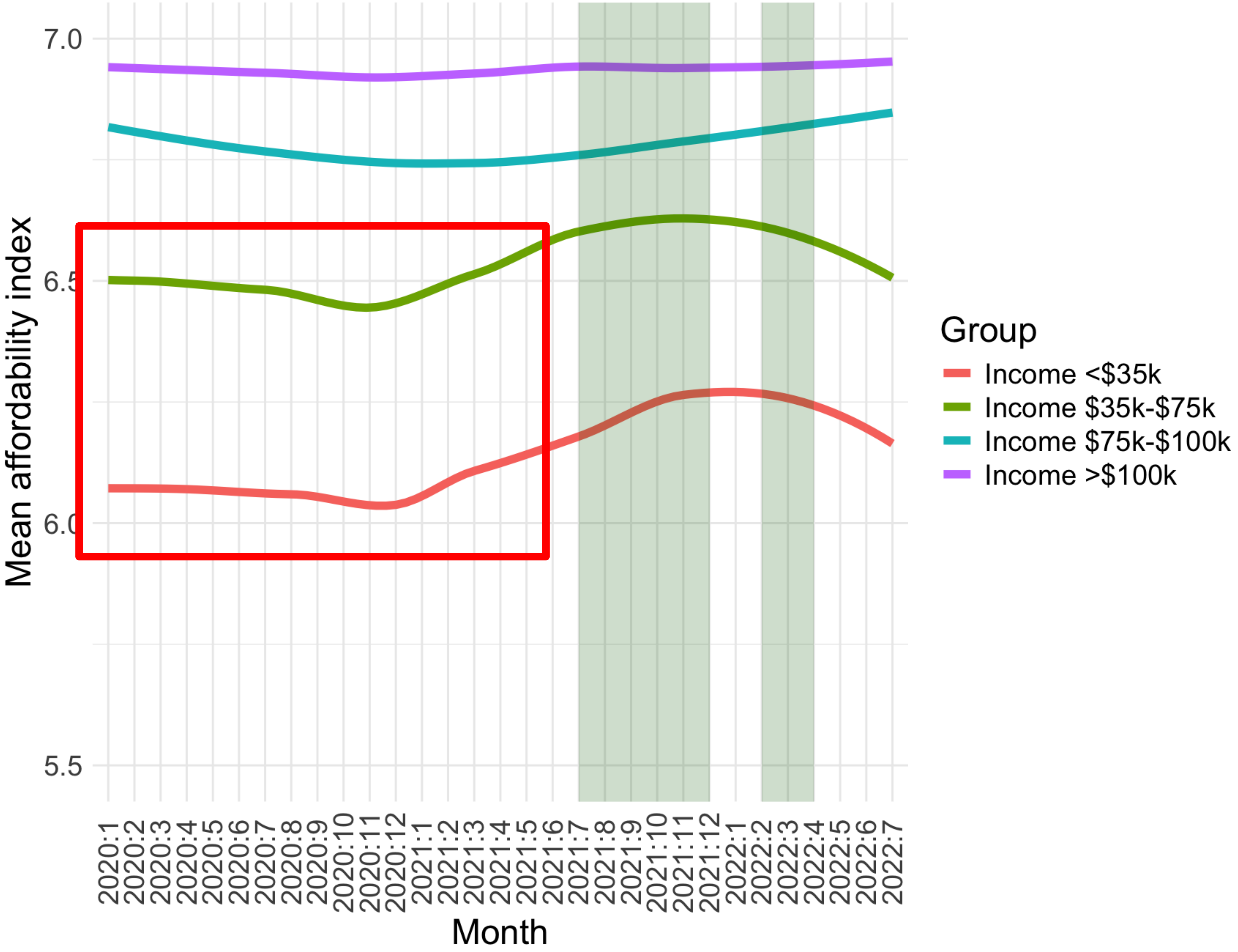
Addressing assumptions in DinD

- **Parallel trends assumption**
  - *Treatment and control would have moved at the same rate in the absence of expanded CTC*



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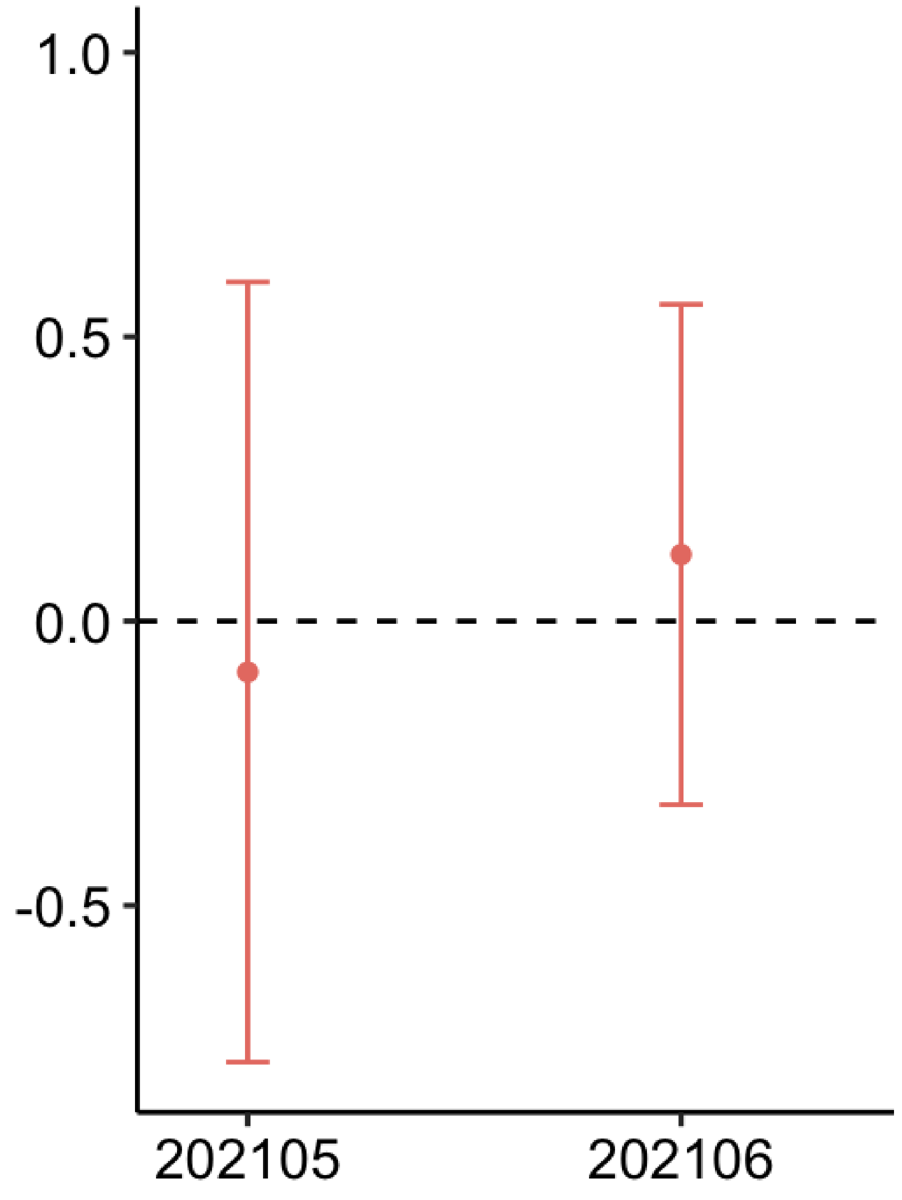
Addressing assumptions in DinD

- **Parallel trends assumption**
  - *Treatment and control would have moved at the same rate in the absence of expanded CTC*
  - Pre-treatment parallel trends assumption tests:
    - Model 1 (during): p = .509
    - Model 2 (persist): p = .804



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Addressing assumptions in DinD

- **Overlap/positivity**
  - *For each treated unit, there must be untreated unit with similar covariates*
  - Partially addressed by comparing narrow income groups (<$35k and $35k-$75k)
  - Accounted for site (clustered standard errors and permutation testing)
- **Absorbing treatment assumption**
  - *Once a unit becomes treated, it stays treated for the rest of the sample*
  - Expanded CTC treatment is not absorbing, instead it switches on and off
  - Split the models to test:
    - *During:* within the July-December 2021 window (treatment is absorbing)
    - *Persistence:* restricted dates to pre-CTC and post-reversion periods only

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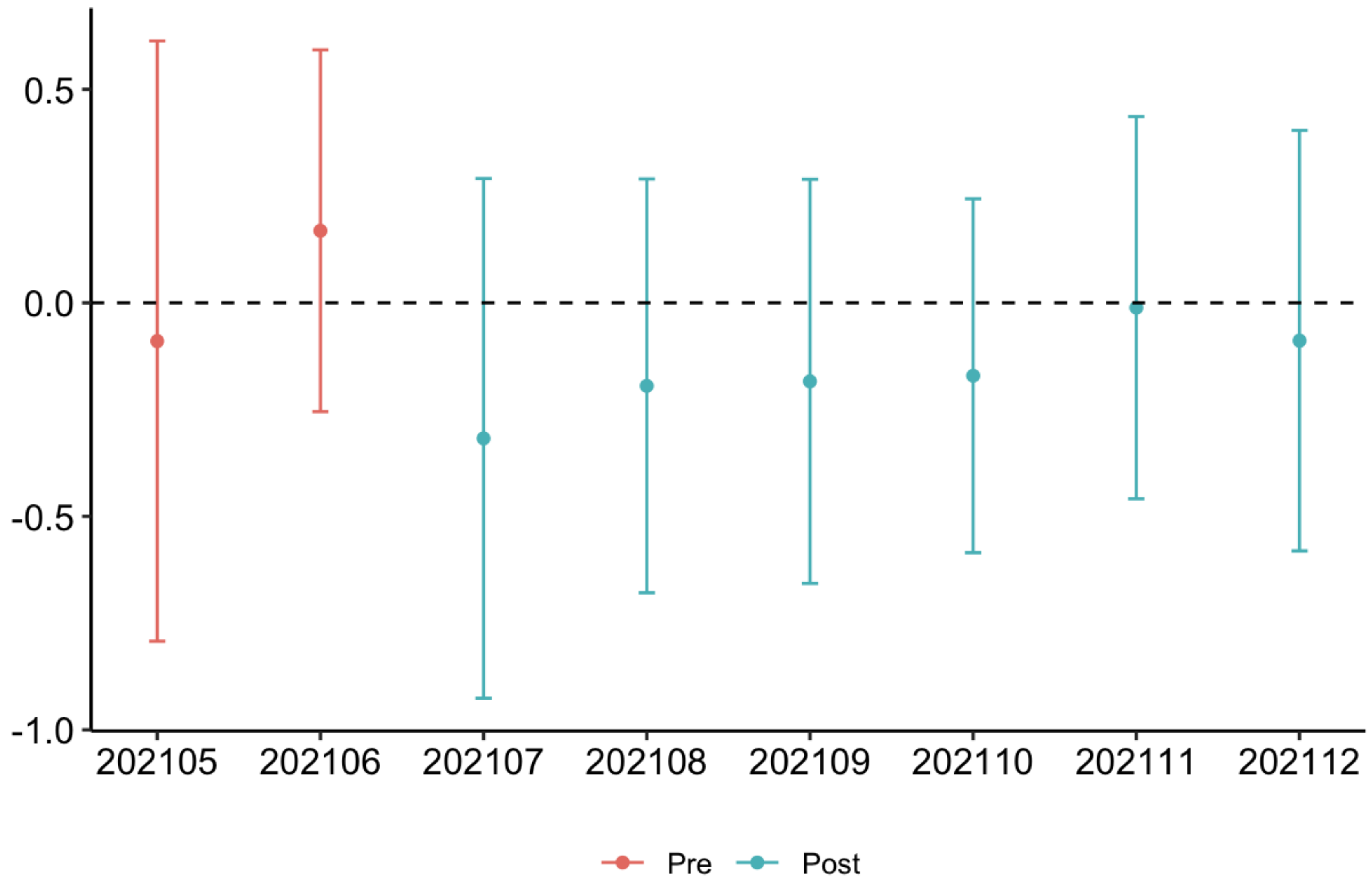
No effect of treatment during the monthly payment window

- Group ATT (controlling for site):
    - -.161, 95% CI [-.39, .07] (*n.s*)

- Affordability got worse for the treated group, but effect was not substantial



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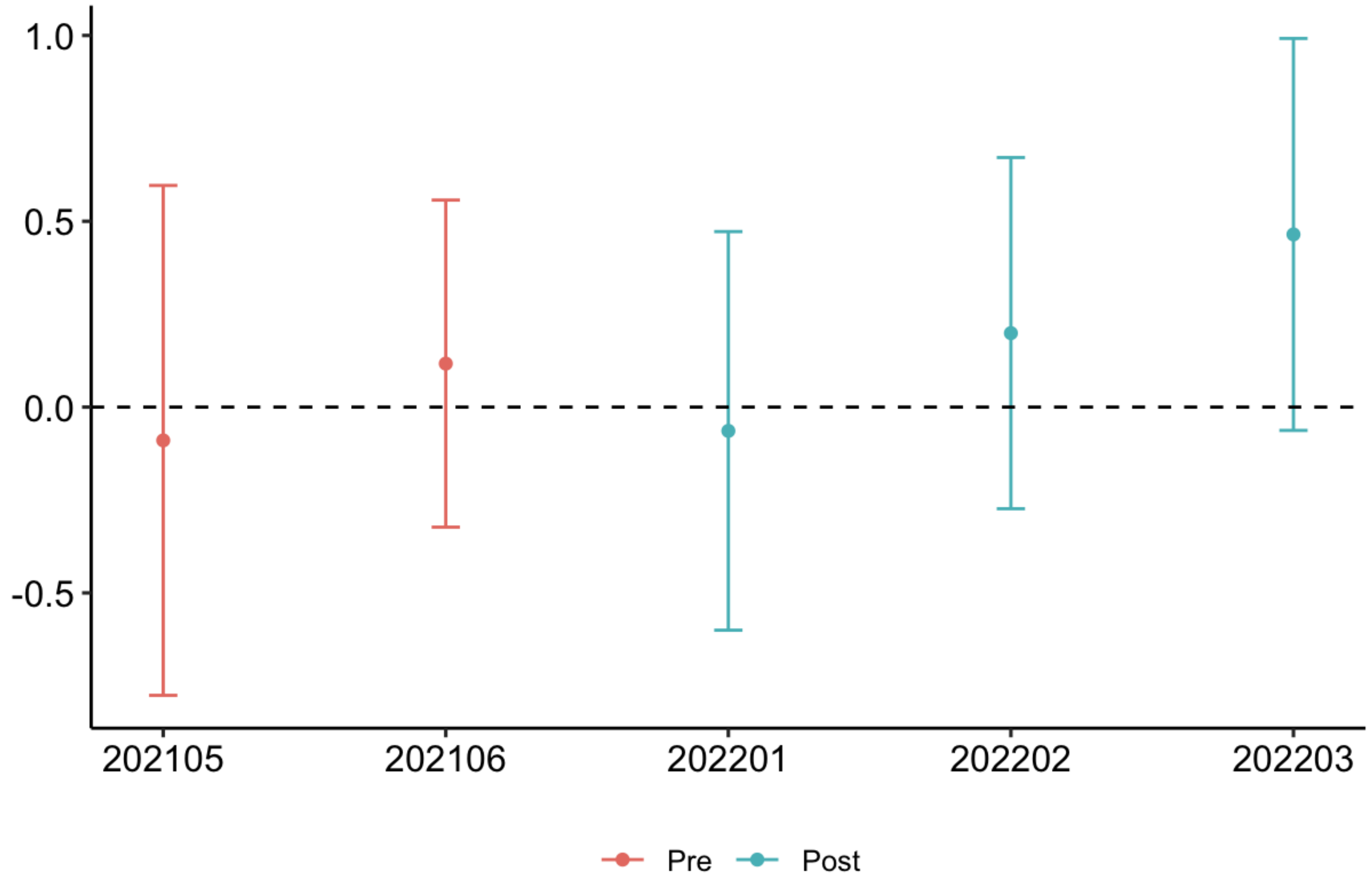
Trend towards improved affordability after monthly payment for lowest income

- Group ATT (controlling for site):
  - .199, 95% CI [-.25, .65], (*n.s)*
- Affordability improved over time for the lowest income group
- The greatest effect was in March 2023
  - .464, 95% CI [-.06, .99], (*n.s.*)



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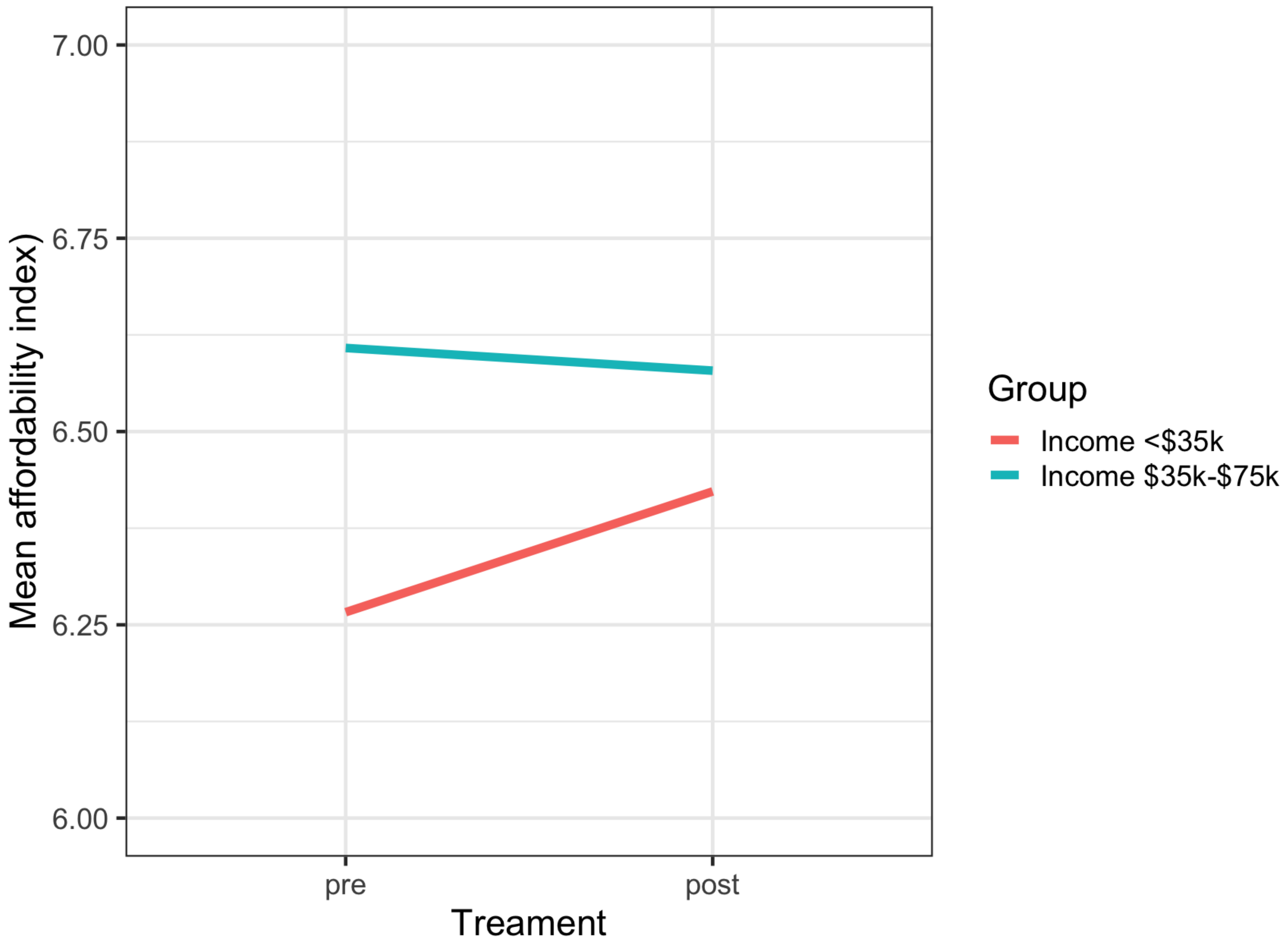

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Summary

- **Strengths:**
  - Capturing impact of policy-based treatment
  - Estimation reflects design structure which reduces bias
  - Converging evidence across robustness checks

- **Limitations:**
  - No direct measure of whether participants definitely received expanded CTC
  - Limited precision in “post” - families filing tax returns at different times

- **Findings/implications**
  - Monthly cash transfer alone did not show effect on affordability for lowest income
  - A persistent improvement for lowest income families coincided with receiving a lump sum (*n.s*)
  - Demonstrates how we can harness rich datasets to:
    - Test impact of policy on development
    - Develop causal designs with observational, developmental data
  - **Next step: mental health outcomes**

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Thank you!

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Data owners:

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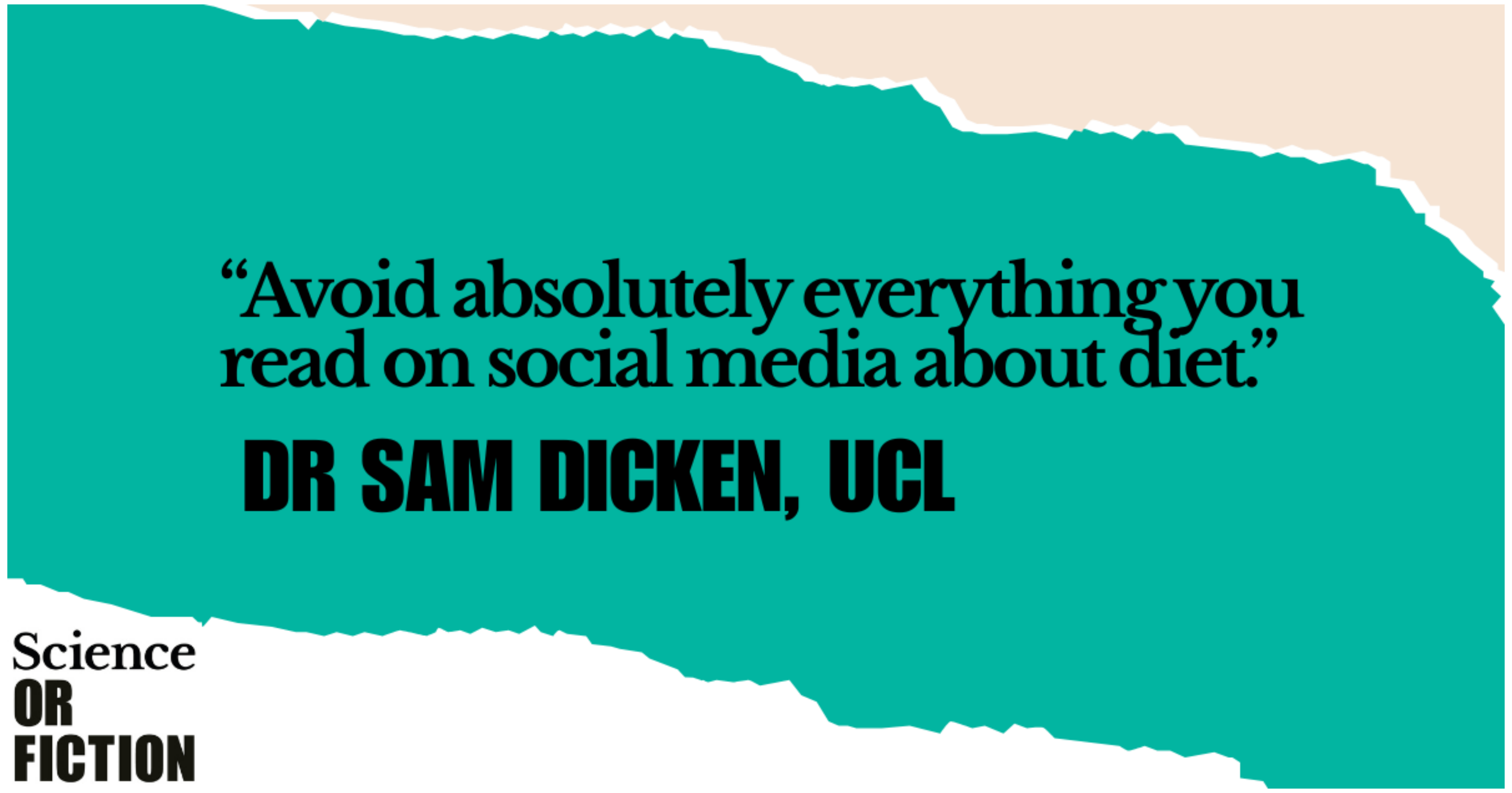

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